NCAA Football Relegation

Economic Simulation Analysis — Sports Analytics Research Project

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R Monte Carlo Linear Regression Random Forest Shiny Dashboard Sports Economics

Overview

This project investigates whether replacing the NCAA Football Bowl Subdivision (FBS) playoff system with a European-style three-tier promotion and relegation structure would accelerate total revenue growth across all 134 FBS programs. A Monte Carlo simulation calibrated from historical NCAA financial data (2015–2024) is supplemented by Linear Regression and Random Forest models to quantify the 10-year revenue impact of structural change in college football.

> **Central finding:** A three-tier relegation system would generate a projected **$10.5B revenue premium** over 10 years compared to the current playoff format, driven by an empirically measured **9.80% engagement premium** from 2015–2024 NCAA data.

Key Results

Model	2025	2034	CAGR	Role
Linear Regression (M5)	$7.5B	$7.5B	0.08%	Feature interpretability
Random Forest (Optimal)	$7.4B	$7.4B	-0.10%	Predictive accuracy
Monte Carlo (Baseline)	$6.0B	$7.6B	2.59%	Status quo simulation
Monte Carlo (Relegation)	**$7.8B**	**$18.1B**	**9.80%**	**Relegation simulation**

- LR Test $R^2 = 0.996$, RMSE = $2.80M per team | RF Test $R^2 = 0.998$, RMSE = $1.91M per team
- Prior year revenue is the single strongest predictor (coef = 0.991, $p < 0.001$)
- LASSO retained only 3 of 12 features — revenue is highly self-perpetuating under the status quo
- RF optimal hyperparameters: 1,200 trees, mtry = 4, min.node.size = 1 (100-combination grid search)
- Relegation expands programs in meaningful competition from 11.9% to 31.3% annually

Data Sources

- USA Today NCAA Athletic Department Revenue Database (primary financials)
- Knight Newhouse College Athletics Database (supplementary financials)
- College Football Reference — win-loss records, rankings, strength of schedule
- 247Sports — recruiting class ratings and composite rankings
- Individual athletic department pages — gap filling for missing programs
- UEFA individual club financial reports — England, Germany, Spain, France, Italy; three tiers per country, 2015–2023
- Transfermarkt — supplementary European club financial data

> **Missing data:** Programs with incomplete financial disclosures were assigned the average revenue of their

> conference positional third (top, middle, bottom) for that year, preserving the known conference-level revenue structure.

Repository Structure

```
collegerelegation/
│
├── cfb_enhanced_part1.R          # Data ingestion and cleaning
├── cfb_enhanced_part2.R          # Feature engineering
├── cfb_enhanced_part3.R          # EDA and parameter extraction
├── DATA_EXTRACTION.R             # Parameter extraction pipeline
├── DATA_DRIVEN_PARAMETERS.R      # Engagement premium calculation
├── PARAMETER_UPDATES.R           # Model configuration
├── FULL_MODEL_V2_PART1.R         # Monte Carlo helper functions
├── FULL_MODEL_V2_PART2.R         # Monte Carlo simulation engine
├── ADD_METRICS.R                 # Gini, mobility, intensity metrics
├── LINEAR_REGRESSION_MODEL.R     # LR with data-driven feature selection
├── RANDOM_FOREST_MODEL.R         # RF with programmatic grid search
├── MODEL_COMPARISON.R            # Cross-model comparison and plots
├── SHINY_DASHBOARD_FINAL.R       # 11-tab interactive dashboard
├── COMPLETE_PIPELINE.R           # Master runner (Phases 1-13)
│
├── ncaa_long.rds                 # Cleaned NCAA data ($M units)
├── euro_long.rds                 # European soccer tier data
├── extracted_parameters.rds      # Historical parameters
├── data_driven_parameters.rds    # Engagement premium and params
├── model_v2_results.rds          # Monte Carlo results
├── lm_results.rds                # Linear Regression results
├── rf_results.rds                # Random Forest results
├── model_comparison_table.rds    # Cross-model comparison
└── all_model_projections.rds     # Combined 2025-2034 projections
```

How to Run

Full Pipeline (Phases 1–13)

```
setwd("C:/Users/lynch/Downloads/collegerelegation")
source("COMPLETE_PIPELINE.R")
```

Run Individual Phases

```
# Phase 9: Linear Regression (~2 min)
source("LINEAR_REGRESSION_MODEL.R")

# Phase 10: Random Forest grid search (~2-3 min)
source("RANDOM_FOREST_MODEL.R")

# Phase 11: Model comparison
source("MODEL_COMPARISON.R")

# Phase 13: Launch Shiny dashboard
source("SHINY_DASHBOARD_FINAL.R")
shinyApp(ui, server)
```

Required R Packages

```
install.packages(c(
  "tidyverse", "ranger", "glmnet", "broom",
  "shiny", "shinydashboard", "DT", "plotly",
  "scales", "ggplot2", "patchwork", "viridis"
```

```
))
```

Model Details

Monte Carlo Simulation

- 100 independent runs per scenario, 5 scenarios total (500 simulations)
- Hierarchical growth: macro AR(1) shock → conference-level rates → team idiosyncratic volatility
- Relegation adds: 9.80% engagement premium, tier multipliers (60% of European levels), 3-year phase-in, revenue floors (65% / 88% / 95% by tier)
- Tier structure: Tier 1 = top 45 programs, Tier 2 = middle 45, Tier 3 = bottom 44; bottom/top 5 swap annually

Linear Regression

- Features ranked by absolute correlation with revenue on training data only
- Five nested OLS specifications (M1: top 1 feature → M5: all 12) plus LASSO at lambda.min and lambda.1se
- 10-fold cross-validation for LASSO lambda selection
- Model selected by holdout test RMSE (2023–2024)
- Train: 3,979 obs (2015–2022) | Test: 1,070 obs (2023–2024)

Random Forest

- 100-combination programmatic grid search derived from p = 13 features and n = 3,979 training rows
- Selection criterion: OOB RMSE (no test data used during tuning)
- Optimal: 1,200 trees, mtry = 4, min.node.size = 1
- Permutation-based variable importance
- Tuning gain: 6.6% lower RMSE, 18.0% lower MAPE vs default specification

Dashboard

An 11-tab interactive Shiny dashboard is included covering:

- Tabs 1–8: Monte Carlo simulation results (Executive Summary, Revenue Dynamics, Competitive Landscape, Equity and Fairness, Conference Impact, Team Explorer, Methodology, Sensitivity Analysis)
- Tab 9: Linear Regression full statistical report (F-statistic, RSE, coefficient table, LASSO feature selection, model comparison)
- Tab 10: Random Forest full report (grid search heatmap, hyperparameter sensitivity, variable importance, partial dependence)
- Tab 11: Cross-model comparison (all four model projections, CAGR comparison, accuracy table)

Deployed at: https://lynchconnor2006.shinyapps.io/collegerelegation

Tier Structure

Tier	Programs	Revenue Floor	Annual Movement
Tier 1	Top 45 by revenue	95% of prior year	Bottom 5 drop to Tier 2
Tier 2	Middle 45 by revenue	88% of prior year	Top 5 promote, bottom 5 drop
Tier 3	Bottom 44 by revenue	65% of prior year	Top 5 promote to Tier 2

Limitations

- Engagement premium measured from revenue rank improvement as a proxy, not direct fan demand or TV ratings data
- European tier multipliers dampened to 60% without a known true NCAA equivalent
- ML models capture the status quo only and cannot model behavioral responses to relegation incentives
- Static tier boundaries do not account for conference realignment or program reclassification

References

- USA Today. (2024). NCAA athletic department revenue database. https://sports.usatoday.com/ncaa/finances
- Sports Reference LLC. (2024). College football statistics and history. https://www.sports-reference.com/cfb
- 247Sports. (2024). College football recruiting rankings. https://247sports.com
- Knight Newhouse Data. (2024). College athletic financial information database. https://knightnewhousedata.org
- UEFA. (2015–2023). Individual club financial reports. https://www.uefa.com
- Transfermarkt. (2024). Club revenue and financial data. https://www.transfermarkt.us